

AWS CLOUD SOLUTION FOR TOMORROW DAY

Cloud Infrastructure Design, Implementation and Infrastructure as Code

Academic Assignment Report

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BSc (Hons) in Information Technology

Specialization: Computer Systems & Network Engineering

Module

AWS Cloud Infrastructure / Cloud Computing Assignment

Case Study

Tomorrow Day – News and Business Information Platform

Technologies & Services

**Amazon VPC • Amazon EC2 • Amazon RDS • Security Groups •
AWS CloudFormation**

Project Scope

AWS Cloud Infrastructure Design, Implementation, Testing & Infrastructure as Code

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1. Introduction

Cloud computing provides organizations with scalable, flexible, and cost-efficient infrastructure without requiring them to maintain large physical data centers. Amazon Web Services (AWS) provides a broad range of cloud services that can be used to design secure, highly available, and scalable application infrastructures.

This assignment focuses on the case of **Tomorrow Day**, a division of News Corp., which provides news and business information services to millions of users worldwide. The company experienced increasing digital traffic in Asia while facing the challenge of an expiring data-center lease in China.

The proposed solution migrates the required infrastructure to AWS using a virtual private cloud, public and private subnets, an Amazon EC2 web server, and an Amazon RDS MySQL database.

The implementation also uses AWS CloudFormation to demonstrate Infrastructure as Code (IaC). The infrastructure is designed to provide internet accessibility to the web tier while keeping the database layer isolated from direct internet access.

2. Case Study Analysis

Tomorrow Day operates a global digital news and business information platform serving millions of users. The company had approximately **12.8 million users in Asia** and generated approximately **90-million-page views per month**.

The company faced several infrastructure challenges:

- The lease for the existing China data center was approaching expiration.
- The Asian digital audience was continuing to grow.
- Physical infrastructure required significant maintenance.
- IT teams spent considerable time managing servers and data centers.
- The company needed lower latency for Asian users.
- Faster deployment of new digital products and services was required.
- The infrastructure is needed to support increasing traffic and business growth.

Cloud-based AWS architecture can address these requirements by providing virtualized infrastructure, flexible scaling, managed database services, network isolation, and Infrastructure as Code.

3. Task 1 – Requirements Identification

3.1 Business Requirements

The following business requirements were identified from the case study:

1. Scalability

The infrastructure must support the existing large Asian user base and accommodate future increases in users and traffic.

2. Reduced Latency

The solution should improve the response time experienced by Asian users by deploying infrastructure in an AWS region geographically closer to the target audience than the previous infrastructure location.

3. Reduced Physical Infrastructure Dependency

The company should reduce its dependency on physical data centers and physical server infrastructure.

4. Reduced Infrastructure Management

IT teams should spend less time maintaining physical servers and data centers and more time focusing on innovation and business activities.

5. **Faster Deployment**

The organization should be able to deploy infrastructure and new services more rapidly.

6. **Improved Reliability**

The cloud solution should provide reliable infrastructure for the company's digital services.

7. **Support for Growing Traffic**

The architecture should be capable of supporting increasing numbers of requests and page views.

8. **Cost Efficiency**

Cloud-based infrastructure should reduce the operational overhead associated with maintaining physical data-center infrastructure.

9. **Security**

Public-facing services and backend database services should be separated using network segmentation and security controls.

10. **Infrastructure Automation**

Infrastructure should be manageable and reproducible through Infrastructure as Code.

3.2 Functional Requirements

The proposed system should provide the following functionality:

- Host the Tomorrow Day web application.
- Allow users to access the web server through the internet.
- Process HTTP requests through the public web server.
- Run the web application using an Amazon EC2 instance.
- Store application data using a relational database.
- Use Amazon RDS MySQL as the database service.
- Allow the EC2 web server to communicate with the RDS database.
- Prevent direct public access to the database.
- Separate public and private resources using subnets.
- Control network traffic using security groups.
- Provide persistent database storage.
- Allow administrators to manage the AWS infrastructure.
- Support infrastructure deployment using AWS CloudFormation.

- Apply appropriate resource tags for identification and management.

3.3 Expected Users

Primary Users

The primary users are the digital customers of Tomorrow Day.

The case study identifies approximately **12.8 million users in Asia**. These users access services such as:

- News articles
- Business information
- Digital content
- Web-based services

Secondary Users

Internal IT / Cloud Operations Team

The IT and cloud operations team is responsible for:

- EC2 management
- VPC management
- Security group configuration
- RDS management
- Infrastructure monitoring
- Infrastructure deployment

System Administrators and Developers

Administrators and developers are responsible for:

- Deploying the web application
- Maintaining application infrastructure
- Managing the database environment
- Troubleshooting application and infrastructure issues

4. Task 2 – Proposed Cloud Architecture

4.1 Architecture Overview

The proposed solution uses AWS to provide a segmented cloud infrastructure consisting of a VPC, public and private subnets, an internet gateway, an EC2 web server, and an Amazon RDS MySQL database.

The web server is deployed in the public subnet because it must be accessible from the internet. The RDS database is deployed in private subnets and does not have public access.

The communication flow is:

Internet Users → Internet Gateway → EC2 Web Server → RDS MySQL Database

The database accepts MySQL traffic only from the web server security group.

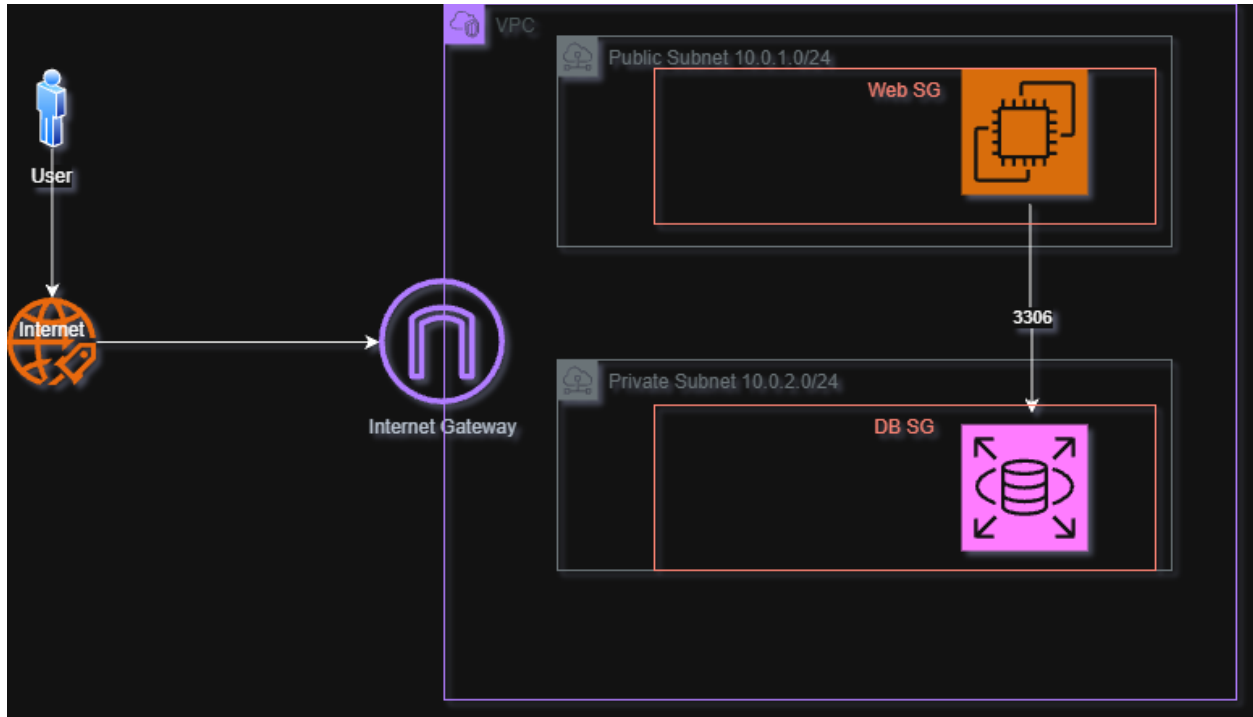


Figure 1 Proposed AWS architecture for the Tomorrow Day cloud solution

The architecture diagram should show:

- Internet / Asian Users
- AWS VPC
- Internet Gateway
- Public Subnet
- EC2 Web Server
- Private Subnet
- RDS MySQL Database
- Web Security Group
- Database Security Group
- HTTP traffic

- MySQL TCP/3306 traffic

4.2 Network Layer

The solution uses an Amazon VPC with the following logical network structure:

Resource	Configuration
VPC	IT20263980_VPC
VPC CIDR	10.0.0.0/16
Public Subnet	IT20263980_PublicSubnet
Public Subnet CIDR	10.0.1.0/24
Private Subnet 1	IT20263980_PrivateSubnet
Private Subnet 1 CIDR	10.0.2.0/24
Private Subnet 2	IT20263980_PrivateSubnet_2
Private Subnet 2 CIDR	10.0.3.0/24
Internet Gateway	IT20263980_IGW
Public Route Table	IT20263980_PublicRT

The public subnet contains the EC2 web server, while the private subnets are used by the RDS database subnet group.

4.3 Compute Layer

Amazon EC2 is used as the compute layer.

The EC2 instance is named:

IT20263980_WebServer

The instance is deployed inside the public subnet and is configured with a public IP address.

Apache HTTP Server is installed on the instance using EC2 User Data.

The web server provides a basic Tomorrow Day web page to verify internet accessibility.

4.4 Storage Layer

Persistent storage is provided through the Amazon RDS database service.

RDS provides managed database storage without requiring the organization to manually maintain database server hardware.

The database storage is associated with the MySQL RDS instance.

4.5 Database Layer

Amazon RDS for MySQL is used as the database layer.

The database is named:

IT20263980_Database

The RDS instance is deployed using a DB subnet group containing private subnets.

The database is configured with:

- MySQL database engine
- Private network placement
- Public access disabled
- TCP port 3306
- Database security group
- Persistent database storage

4.6 Security Architecture

Security is implemented using network segmentation and security groups.

Web Security Group

The web security group allows:

Protocol	Port	Source
HTTP	80	0.0.0.0/0
SSH	22	Administrator's IP
Outbound	All	0.0.0.0/0

HTTP is publicly accessible because the web server must serve internet users.

SSH access is restricted to the administrator's IP address.

Database Security Group

The database security group allows:

Protocol	Port	Source
MySQL	3306	IT20263980_WebSG
Outbound	All	0.0.0.0/0

The database does not accept MySQL connections directly from the public internet.

5. Task 3 – AWS Cloud Implementation

5.1 AWS Environment

The infrastructure was implemented using the **AWS Academy Sandbox Environment**.

The following AWS services were used:

- Amazon VPC
- Internet Gateway
- Amazon EC2
- Amazon RDS
- Security Groups
- Route Tables
- Subnets
- AWS CloudFormation

The resources were named using the required Student ID naming convention.

5.2 VPC Configuration

A dedicated VPC was created for the Tomorrow Day solution.

VPC Name: IT20263980_VPC

CIDR: 10.0.0.0/16

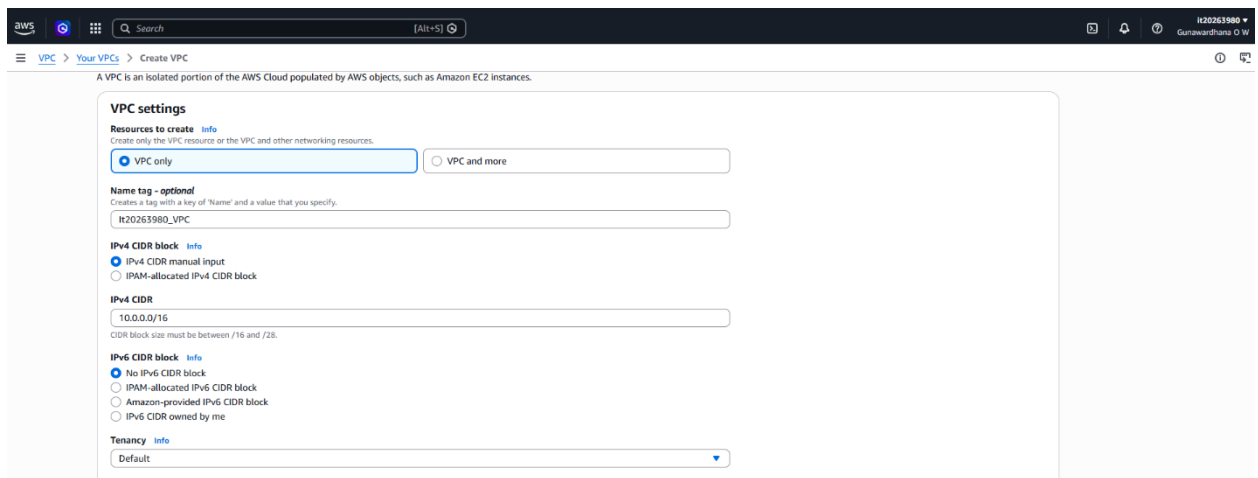
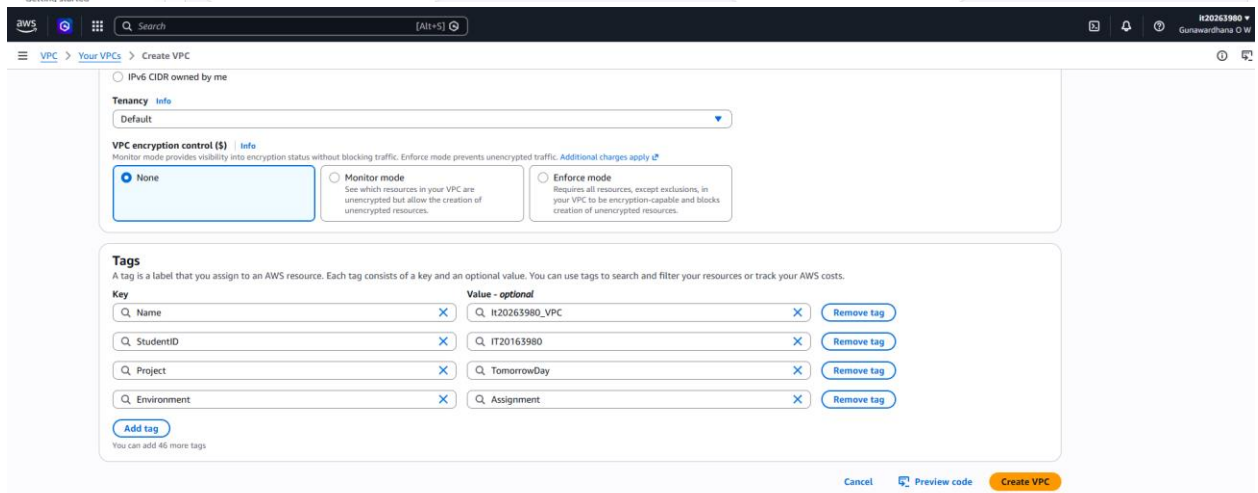
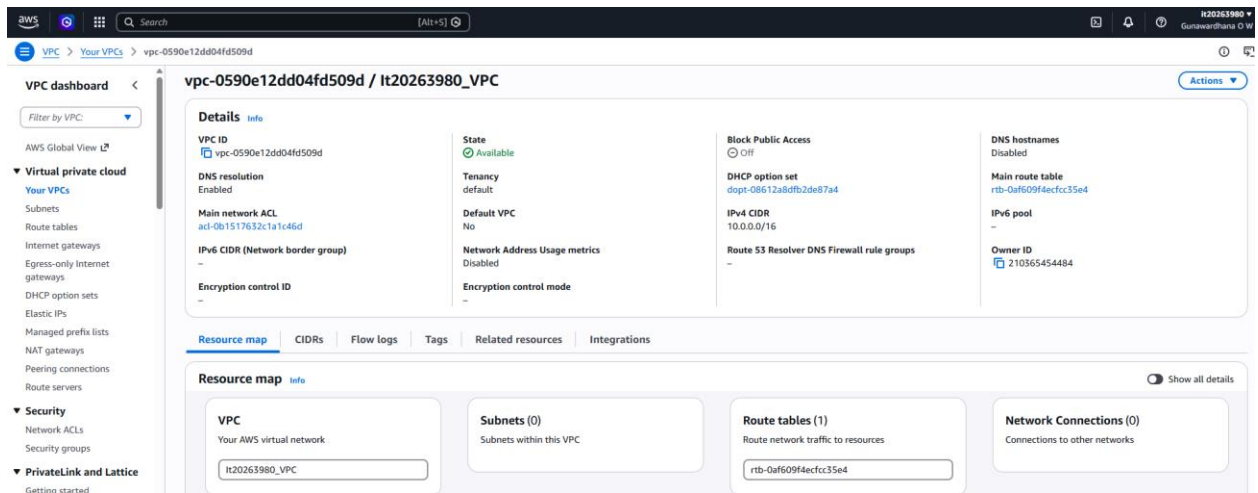


Figure 2 VPC Details – IT20263980_VPC

These screenshots demonstrate the VPC configuration and details.

5.3 Subnet Configuration

Three subnets were used in the proposed infrastructure.

Public Subnet

Name: IT20263980_PublicSubnet

CIDR: 10.0.1.0/24

The public subnet is associated with the public route table and provides network connectivity required by the web server.

Private Subnet 1

Name: IT20263980_PrivateSubnet

CIDR: 10.0.2.0/24

Private Subnet 2

Name: IT20263980_PrivateSubnet_2

CIDR: 10.0.3.0/24

The two private subnets are used by the RDS DB subnet group.

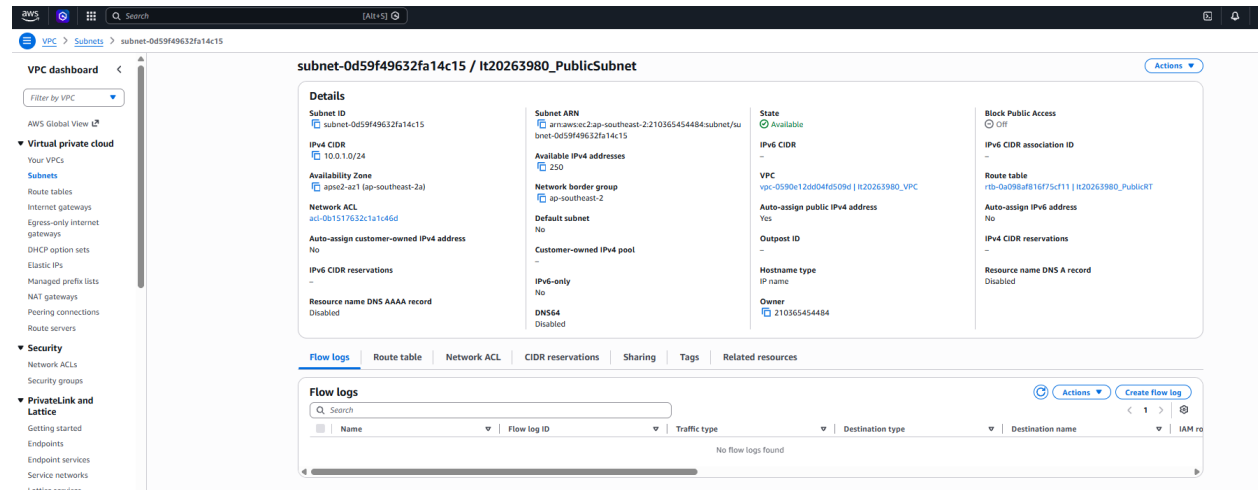


Figure 3 EC2 deployed in the public subnet

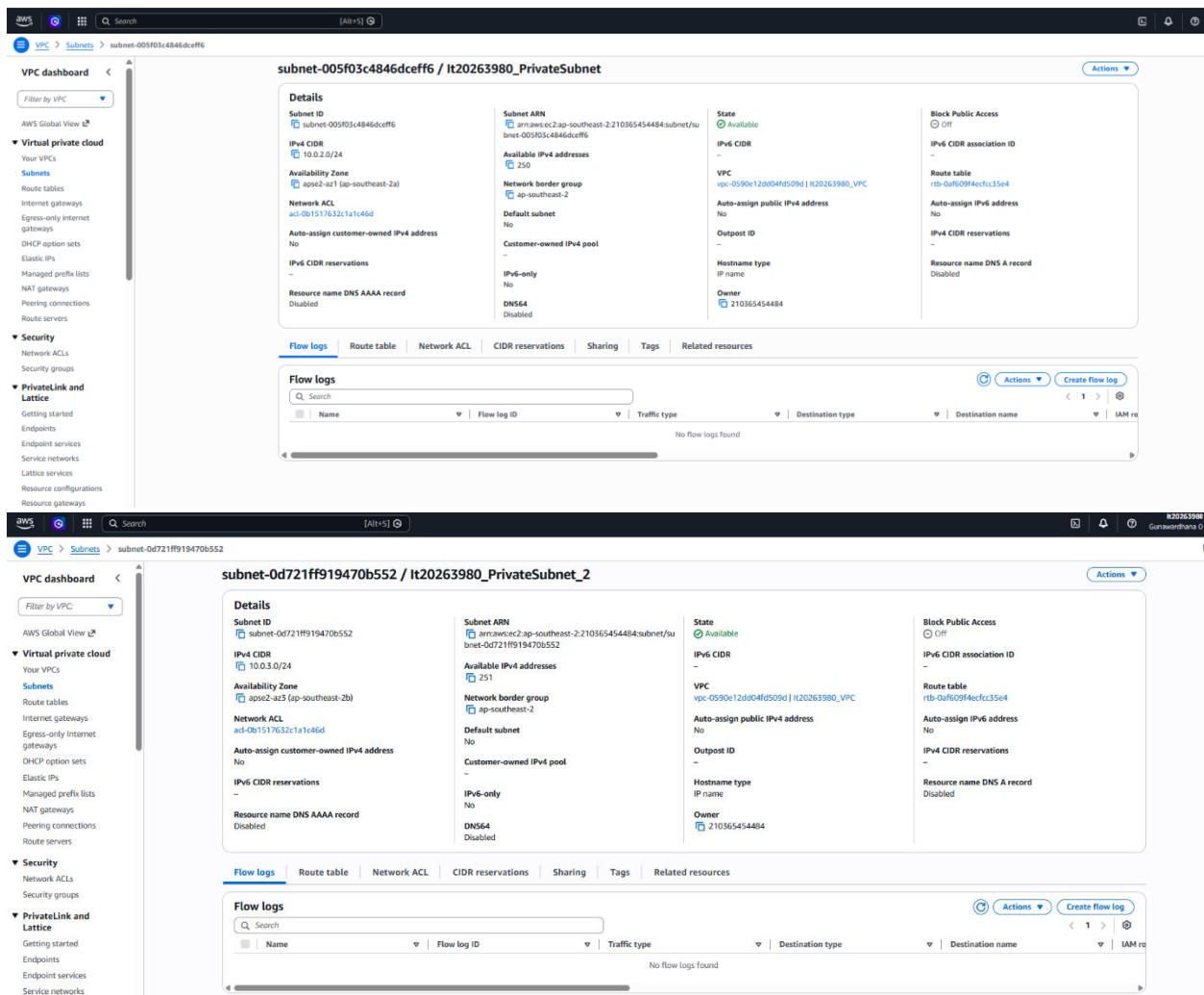


Figure 4 RDS deployed using private subnets

5.4 Internet Gateway

An Internet Gateway was created and attached to the VPC.

Name: IT20263980_IGW

The Internet Gateway provides internet connectivity to resources in the public subnet through the associated route table.

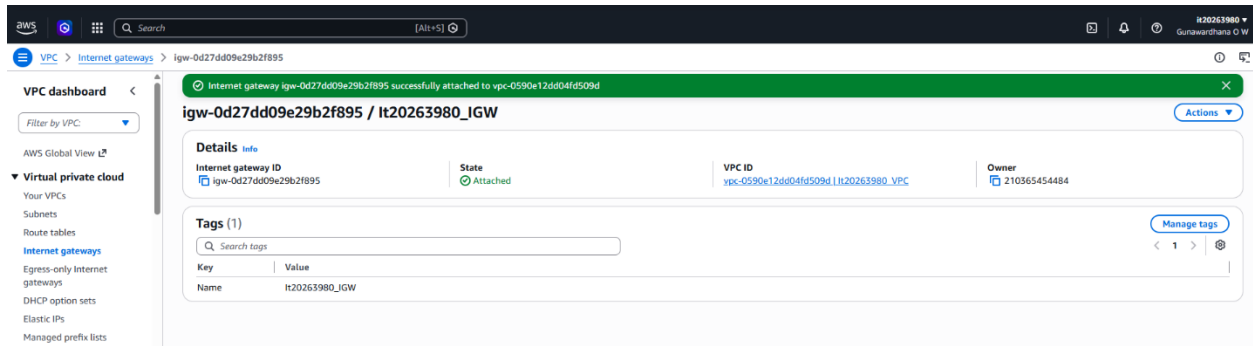


Figure 5 Internet Gateway configuration

5.5 Route Table

A public route table was created for the web server subnet.

Name: IT20263980_PublicRT

The route table contains the following internet route:

Destination: 0.0.0.0/0

Target: IT20263980_IGW

The public subnet is associated with this route table.

This configuration enables the EC2 web server to receive internet traffic.

5.6 Web Security Group

The web server security group was created as:

IT20263980_WebSG

The security group controls inbound and outbound traffic for the EC2 web server.

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

It20263980_Web5G

Name cannot be edited after creation.

Description [Info](#)

Security group for Tomorrow Day web server

VPC [Info](#)

vpc-0590e12dd04fd509d (It20263980_VPC)

Inbound rules [Info](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#) Source [Info](#) Description - optional [Info](#)

HTTP TCP 80 Anywhere... 0.0.0.0/0 X Delete

SSH TCP 22 My IP 112.135.72.77/32 X Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#) Destination [Info](#) Description - optional [Info](#)

All traffic All All Custom 0.0.0.0/0 X Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Create security group

Figure 6 Web server security group

Security Group Rules

VPC > Security Groups > sg-0b30725a002d6f04f

VPC dashboard < Filter by VPC AWS Global View

Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only internet gateways DHCP option sets Elastic IPs Managed prefix lists NAT gateways Peering connections Route servers Security Network ACLs

sg-0b30725a002d6f04f - It20263980_WebSG Actions

Details

Security group name It20263980_Web5G

Owner 210365454484

Security group ID sg-0b30725a002d6f04f

Inbound rules count 2 Permission entries

Description Security group for Tomorrow Day web server

VPC ID vpc-0590e12dd04fd509d

Outbound rules count 1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0fdef2bc371099cd6	IPv4	SSH	TCP	22	112.135.72.77/32	-
☐	-	sgr-0bca5d56611a8ebac	IPv4	HTTP	TCP	80	0.0.0.0/0	-

Figure 7 Web Security Group Rules

The HTTP rule allows users to access the web server from the internet, while SSH access is restricted to the administrator's IP address.

5.7 EC2 Web Server

An Amazon EC2 instance was launched using an Amazon Linux image.

Instance Name: IT20263980_WebServer

The EC2 instance was configured in:

- VPC: IT20263980_VPC
- Subnet: IT20263980_PublicSubnet
- Public IP: Enabled
- Security Group: IT20263980_WebSG

Apache HTTP Server was installed using EC2 User Data.

The web page was configured to display:

Tomorrow Day – AWS Cloud Solution

and the Student ID.

Web Server Creation Screenshot

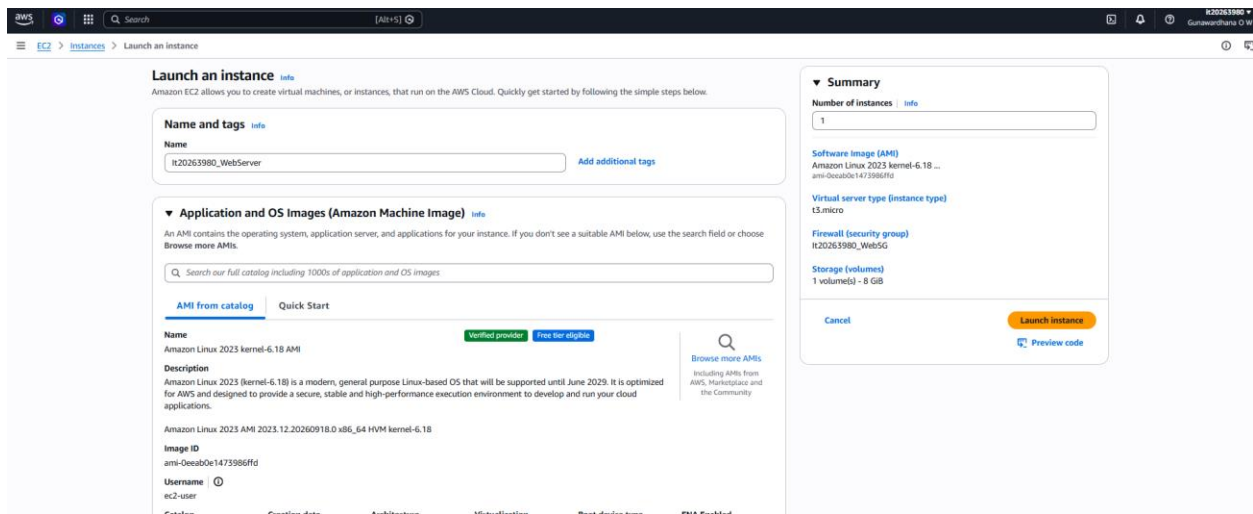


Figure 8 EC2 Web Server creation

EC2 Details Screenshot

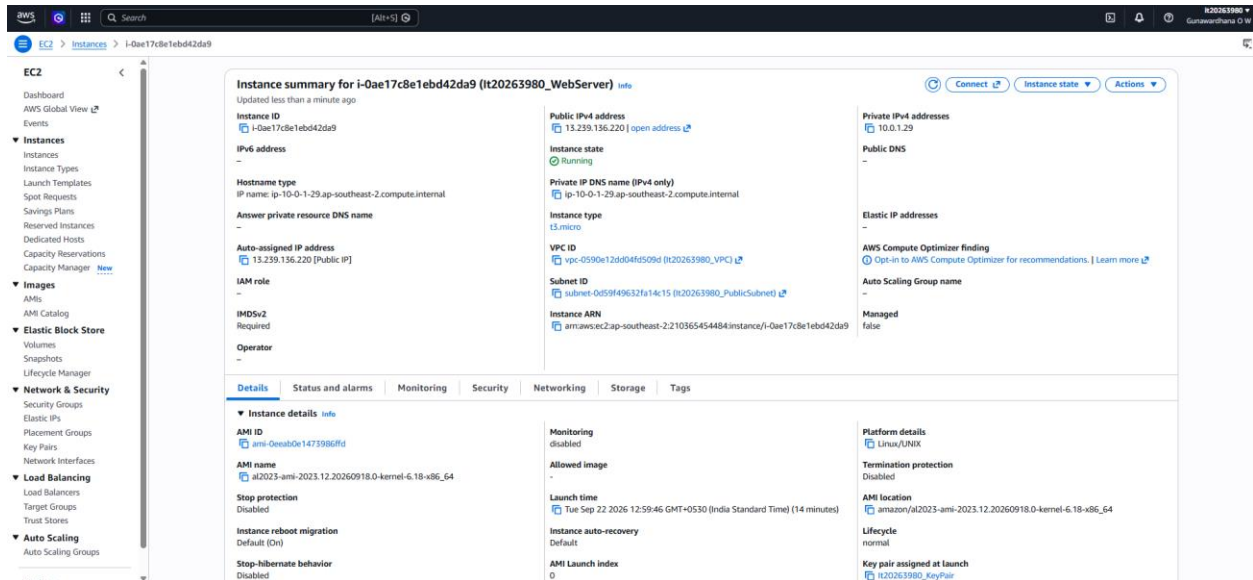
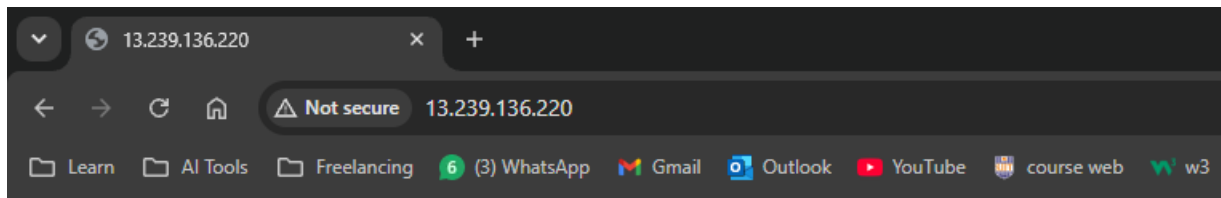


Figure 9 EC2 Web Server details

Internet Web Server Test



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Student ID: IT20263980

Web Server: Amazon EC2

Figure 10 Tomorrow Day web server accessed from the internet

This confirms that the EC2 web server is accessible through the internet.

5.8 RDS MySQL Database

Amazon RDS for MySQL was provisioned as the database layer.

Database Identifier: IT20263980_Database

The database was configured inside the VPC using a DB subnet group containing private subnets.

Public access was disabled to prevent direct internet connectivity to the database.

RDS Details

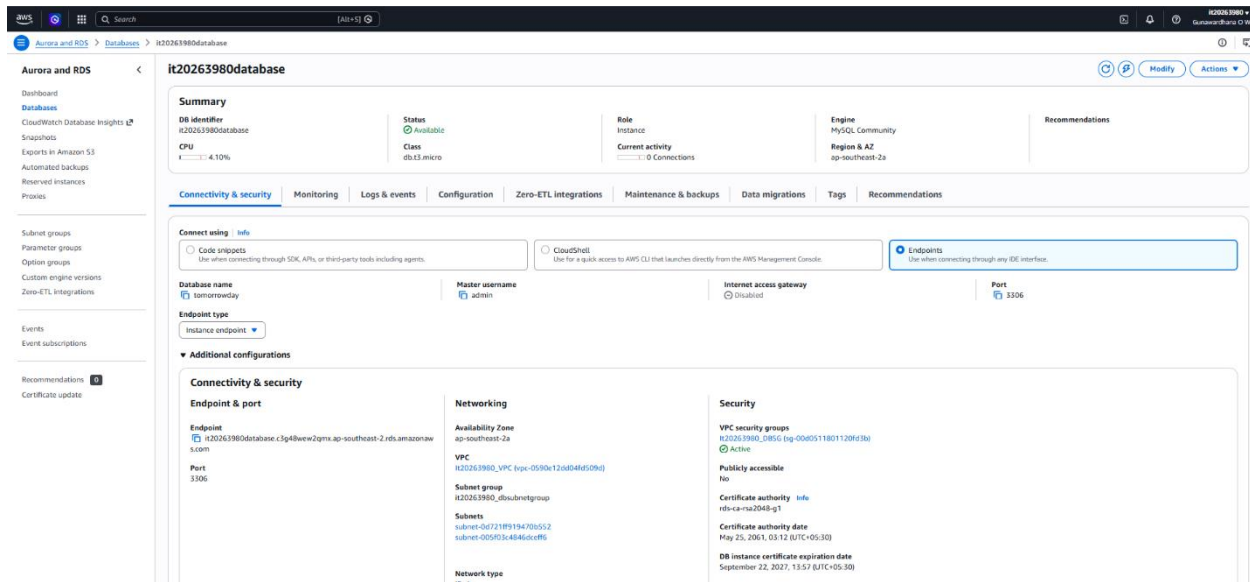


Figure 11 RDS MySQL database details

5.9 Database Security Group

The database security group is:

IT20263980_DB-SG

The security group permits MySQL traffic only from the web server security group.

Screenshot

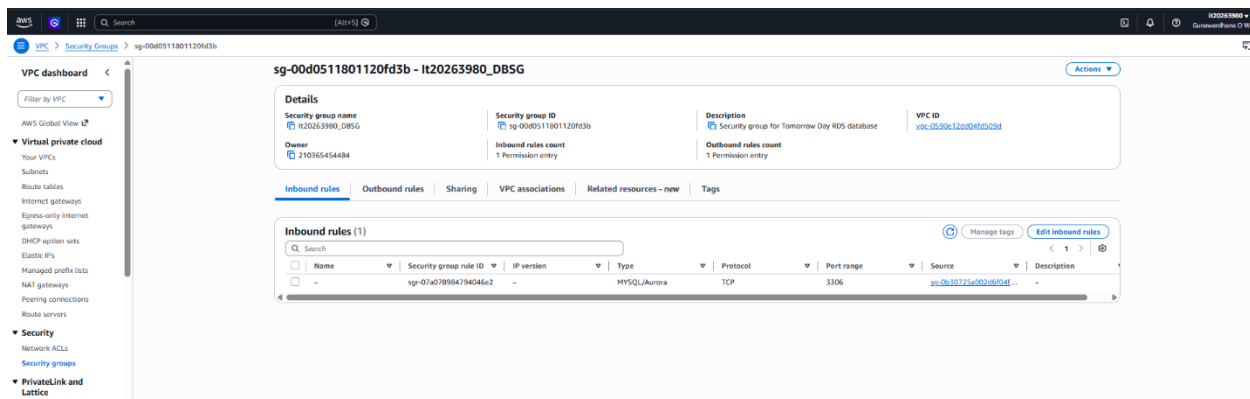


Figure 12 RDS Database Security Group

Database Security Group Rules

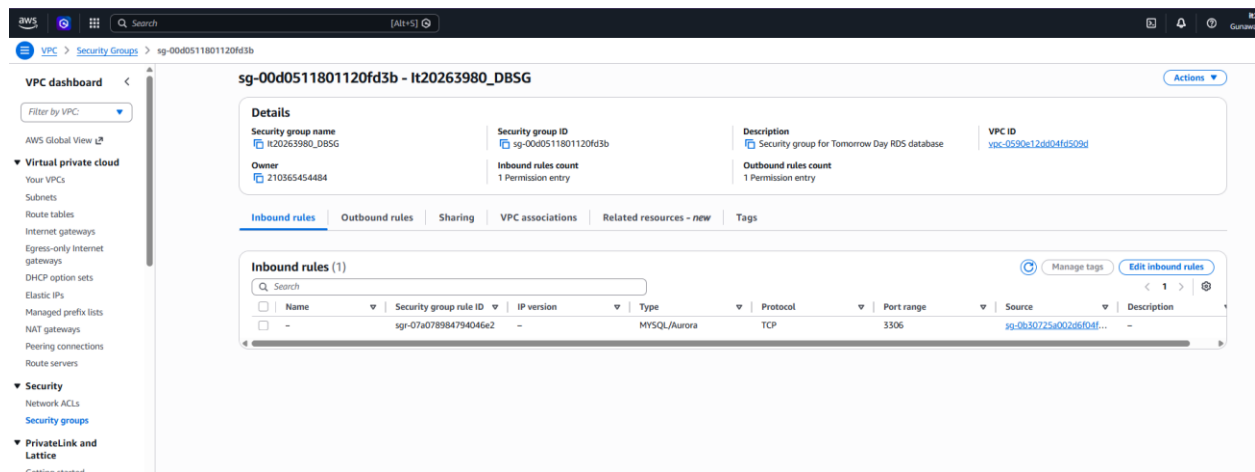


Figure 13 Database Security Group Rules

The important security configuration is:

TCP 3306 → Source: IT20263980_WebSG

This prevents unrestricted internet access to the database.

6. Task 4 – Infrastructure as Code Using CloudFormation

6.1 CloudFormation Overview

AWS CloudFormation was used to implement the infrastructure using Infrastructure as Code.

Instead of manually creating each AWS resource, the required infrastructure can be defined in a YAML template and deployed as a CloudFormation stack.

The CloudFormation template was written specifically for the Tomorrow Day infrastructure.

The template includes resources such as:

- VPC
- Internet Gateway
- Public Subnet
- Private Subnet
- Private Subnet 2
- Route Table
- Route Table Association
- Web Security Group
- Database Security Group

- EC2 Instance
- RDS DB Subnet Group
- RDS MySQL Database

The Student ID is passed as a parameter and is used for resource naming.

6.2 CloudFormation Template Design

The CloudFormation template follows a parameterized design.

The Student ID parameter allows resource names to follow the required naming convention.

Example naming structure:

IT20263980_VPC

IT20263980_WebServer

IT20263980_Database

IT20263980_WebSG

IT20263980_DB-SG

Appropriate tags are also applied to resources.

The main tags used are:

- StudentID = IT20263980
- Project = TomorrowDay
- Environment = Assignment
- Owner = D.Heshan.Lakshitha

6.3 CloudFormation Stack Deployment

The CloudFormation stack was created using the AWS Management Console.

Stack Name:

IT20263980_CloudFormation

CloudFormation Stack Creation

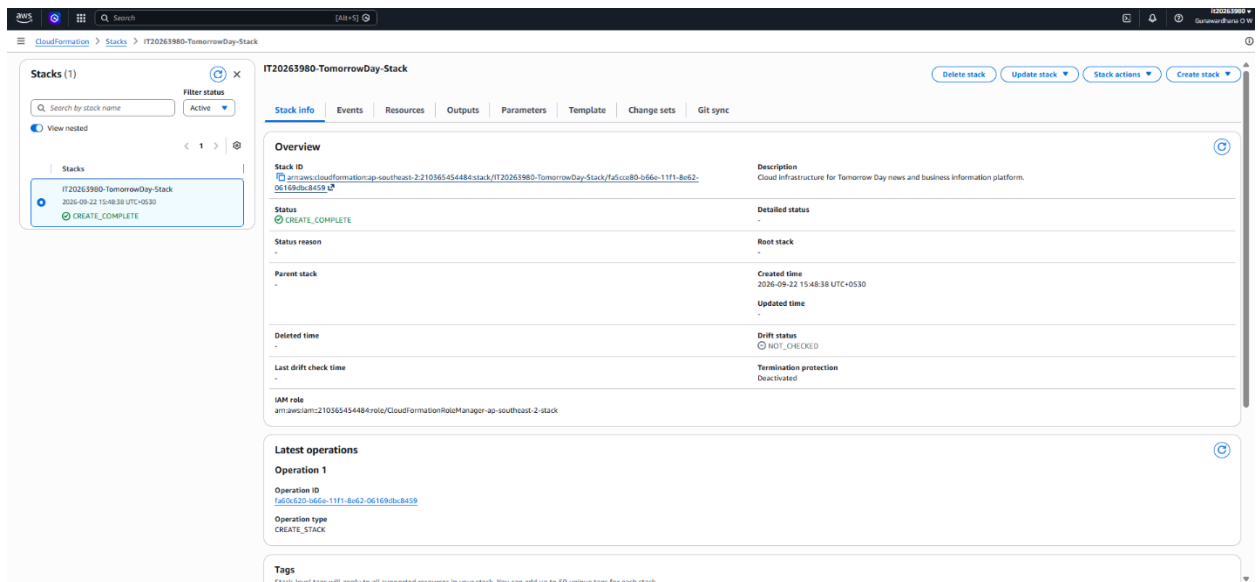


Figure 14 CloudFormation stack creation

CloudFormation Parameters

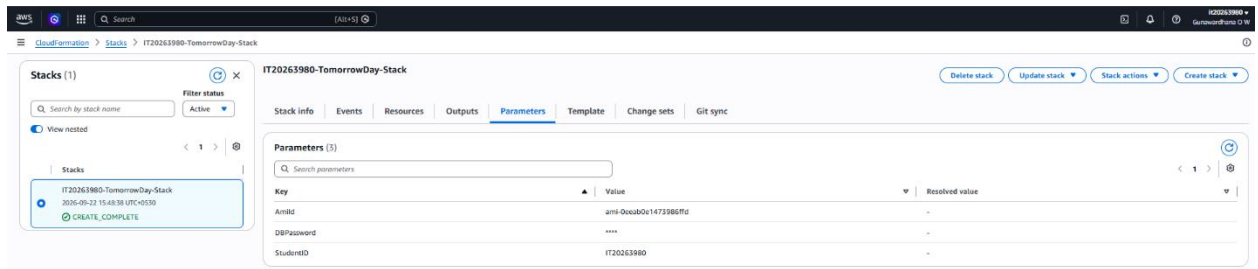


Figure 15 CloudFormation Student ID parameter

Successful Stack Creation

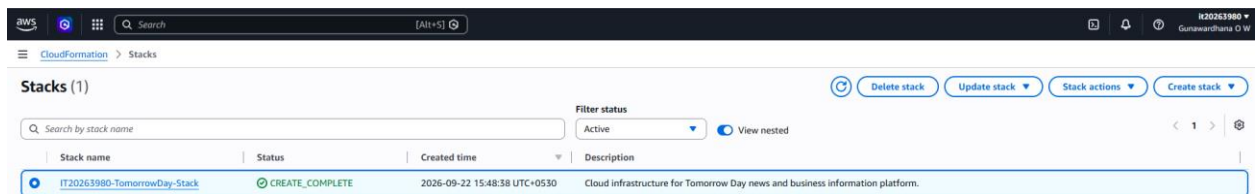


Figure 16 CloudFormation stack CREATE_COMPLETE status

Stack Creation Complete

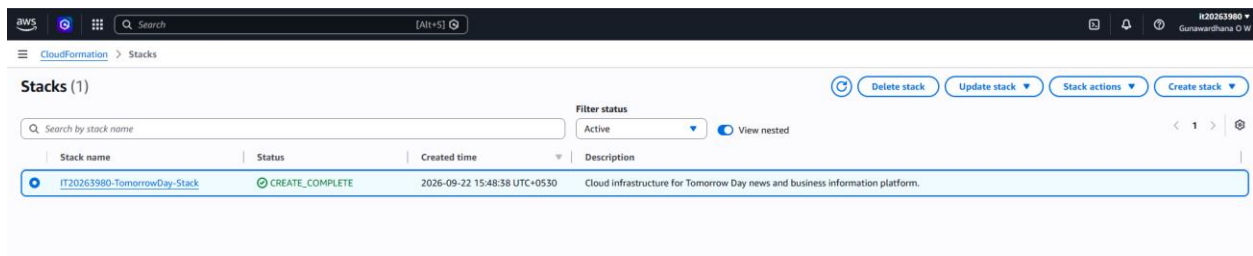


Figure 17 Successfully created CloudFormation stack

6.4 CloudFormation Resources

After successful deployment, CloudFormation created the resources defined in the template.

The resources can be viewed under the **Resources** section of the CloudFormation stack.

Screenshot

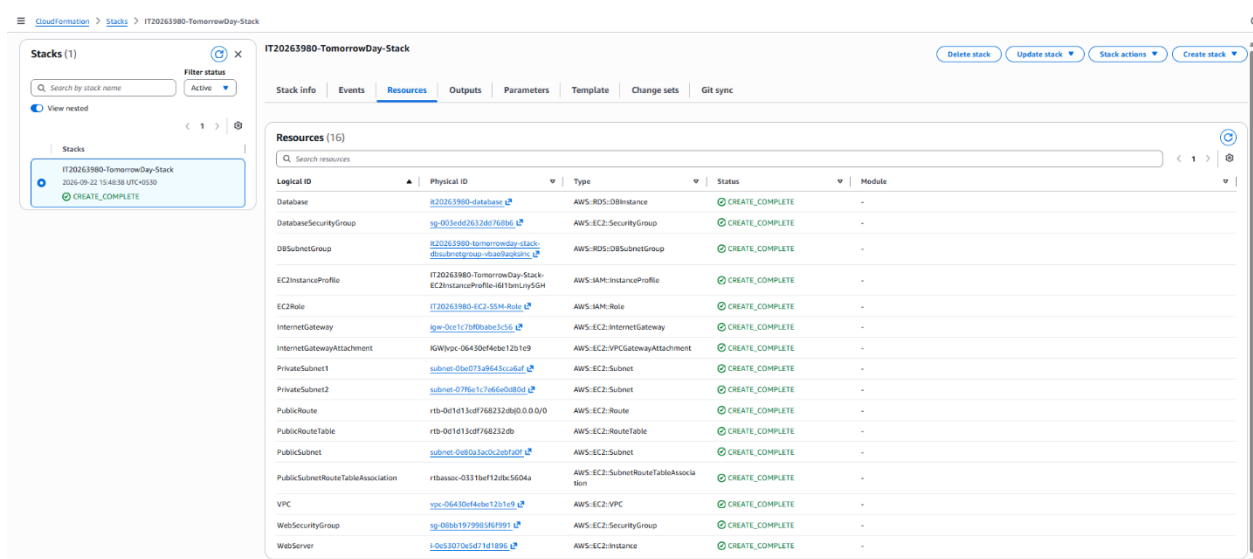


Figure 18 CloudFormation-created resources

The resource list provides evidence that the infrastructure was successfully provisioned using Infrastructure as Code.

6.5 CloudFormation Outputs

The CloudFormation stack outputs provide important information about the deployed infrastructure.

Screenshot

Key	Value	Description	Export name
DatabaseEndpoint	r-20263980-database-c9a8wev2ms.ap-southeast-2.rds.amazonaws.com	RDS database endpoint	-
DatabasePort	3306	RDS database port	-
DatabaseSecurityGroup	sg-059eb26328076896	Database security group ID	-
PrivateSubnet1Id	subnet-0be073a9645c0a8f	Private subnet 1 ID	-
PrivateSubnet2Id	subnet-07661c706e0d080d	Private subnet 2 ID	-
PublicSubnetId	subnet-0e80a3ac0c2ebfa0f	Public subnet ID	-
VPCId	vpc-06430e40be12b1e9	VPC ID	-
WebSecurityGroup	sg-08bb19798636f991	Web server security group ID	-
WebServerInstance	i-0e53070e5d71d1896	EC2 Web Server Instance ID	-
WebServerPublicIP	54.79.54.169	Public IP address of web server	-

Figure 19 CloudFormation stack outputs

6.6 CloudFormation-Created EC2 Instance

The EC2 instance created by the CloudFormation stack was verified through the EC2 console.

Screenshot

Instance summary for i-0e53070e5d71d1896 (IT20263980_WebServer)

Updated less than a minute ago

Instance ID
i-0e53070e5d71d1896

IPv4 address
-

Hostname type
IP name: ip-10-0-1-118.ap-southeast-2.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
54.79.54.169 (Public IP)

IAM role
IT20263980-EC2-SSH-Role

IMDSv2
Required

Operator
-

Public IPv4 address
54.79.54.169 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-0-1-118.ap-southeast-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-06430e40be12b1e9 (IT20263980_VPC)

Subnet ID
subnet-0e80a3ac0c2ebfa0f (IT20263980_PublicSubnet)

Instance ARN
arn:aws:ec2:ap-southeast-2:210365454484:instance/i-0e53070e5d71d1896

Private IPv4 addresses
10.0.1.118

Public DNS
ec2-54-79-54-169.ap-southeast-2.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer Finding
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

Auto Scaling Group name
-

Managed
False

Instance details

AMI ID
ami-0eab0e14739860f1

AMI name
al2023-ami-2023.12.20206918.0-kernel-6.18-x86_64

Stop protection
Disabled

Monitoring
disabled

Allowed image
-

Launch time
Tue Nov 27 20:26:15 UTC+10 (GMT+10:00) (Brisbane Standard Time) (6 minutes)

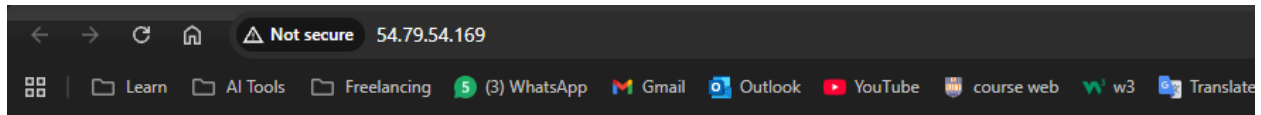
Platform details
Linux/UNIX

Termination protection
Disabled

AMI location
amazon2023.12.20206918.0-kernel-6.18-x86_64

Figure 20 EC2 instance created through CloudFormation

CloudFormation Web Server Test



Tomorrow Day - AWS Cloud Solution

Web Server Successfully Deployed

Student ID: IT20263980

Server: Amazon EC2

Deployment: AWS CloudFormation

Figure 21 CloudFormation-created web server accessibility test

6.7 CloudFormation-Created RDS Database

The RDS database provisioned by the CloudFormation stack was verified through Amazon RDS.

Screenshot

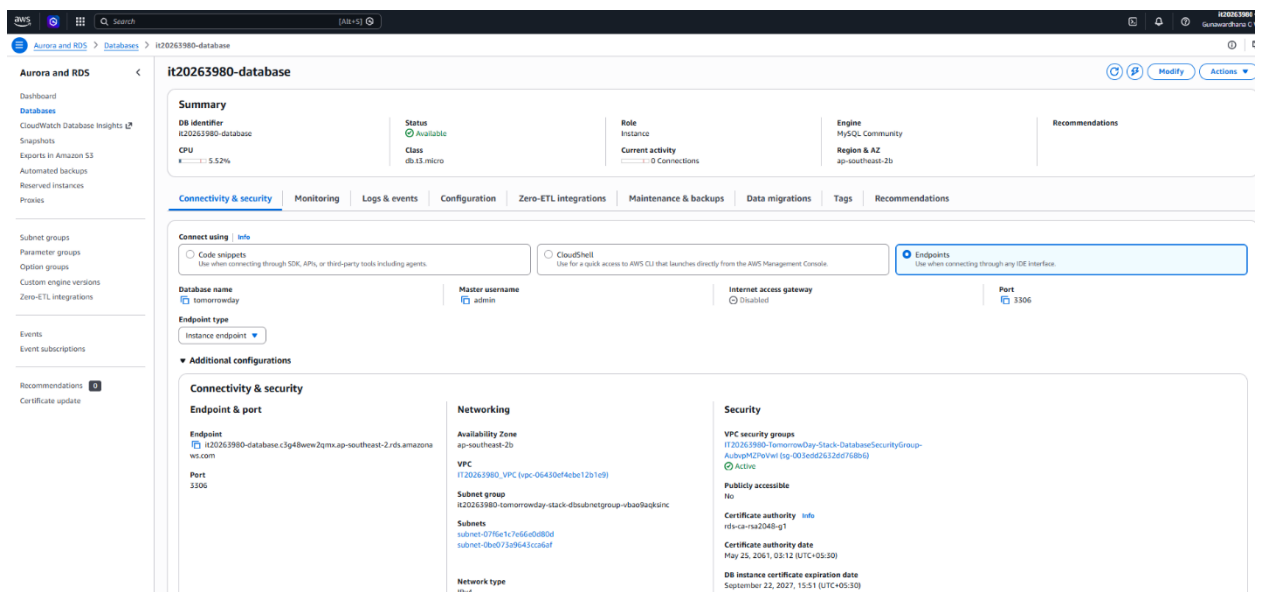


Figure 22 RDS database created through CloudFormation

CloudFormation Database Security Group

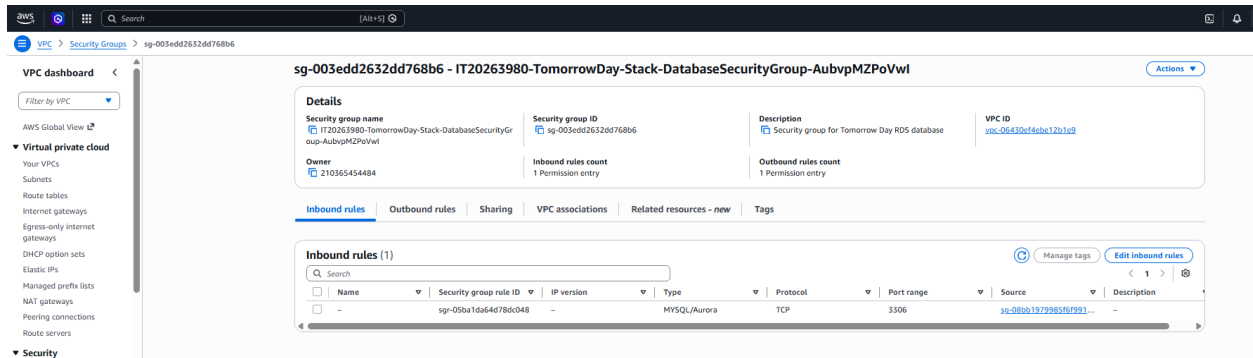


Figure 23 Database security group created through CloudFormation

7. Task 5 – Testing and Validation

The deployed infrastructure was tested to validate the major functional and security requirements.

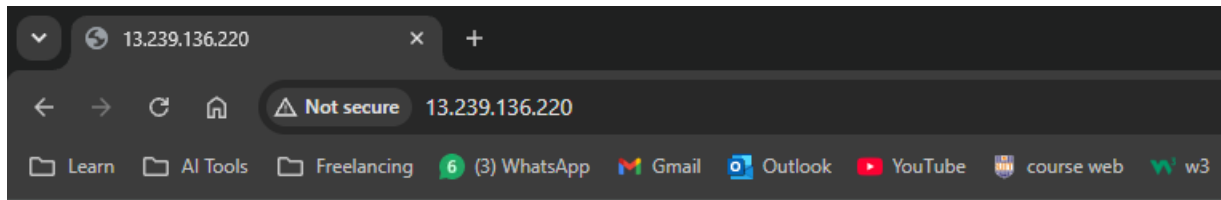
7.1 Web Server Accessibility from the Internet

The EC2 web server was accessed through its public IP address using a web browser.

The successful response confirms that:

- EC2 is deployed in the public subnet.
- The instance has internet connectivity.
- The Internet Gateway is correctly configured.
- The public route is functioning.
- HTTP port 80 is allowed by the Web Security Group.
- Apache HTTP Server is running.

Screenshot



Tomorrow Day - AWS Cloud Solution

Student ID: IT20263980

Web Server: Amazon EC2

Figure 24 Web server accessibility from the internet

7.2 EC2 to RDS Connectivity

The RDS database was tested from the EC2 web server.

The database connection uses:

Protocol: MySQL

Port: 3306

Source: EC2 Web Server

A successful connection demonstrates that the web server can communicate with the RDS database.

The connection path is:

EC2 Web Server → TCP 3306 → RDS MySQL

Screenshot

```
Session ID: d4d8448-2d61-7002-8bb8-724fd67652-b6ca47hq7eq7y6d4+@2u Shortcuts Instance ID: i-0e530706d76189618720261980_WebServer Terminate session

Running scriptlet: mariadb185-3:10.5.29-1.amzn2023.0.1.x86_64 5/5
Verifying : mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64 1/5
Verifying : mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch 2/5
Verifying : mariadb185-3:10.5.29-1.amzn2023.0.1.x86_64 3/5
Verifying : mariadb185-common-3:10.5.29-1.amzn2023.0.1.x86_64 4/5
Verifying : perl-Sys-Hotname-1.23-477.amzn2023.0.9.x86_64 5/5

Installed:
mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64 mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch mariadb185-3:10.5.29-1.amzn2023.0.1.x86_64 mariadb185-common-3:10.5.29-1.amzn2023.0.1.x86_64 perl-Sys-Hotname-1.23-477.amzn2023.0.9.x86_64

Complete!
sh-5.25 mysql > h (120263980-database.c3g0hw02ms.ap-southeast-2.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MySQL connection id is 32
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| tomorrowday |
+-----+
5 rows in set (0.005 sec)

MySQL [(none)]> USE tomorrowday;
Database changed
MySQL [tomorrowday]>
MySQL [tomorrowday]> CREATE TABLE test_table (
-> id INT PRIMARY KEY AUTO_INCREMENT,
-> message VARCHAR(255)
-> );
Query OK, 0 rows affected (0.183 sec)

MySQL [tomorrowday]>
MySQL [tomorrowday]> INSERT INTO test_table (message)
-> VALUES ('Tomorrow Day CloudFormation Test');
Query OK, 1 row affected (0.006 sec)

MySQL [tomorrowday]>
MySQL [tomorrowday]> SELECT * FROM test_table;
+----+ message +-----+
| 1 | Tomorrow Day CloudFormation Test |
+----+-----+
1 row in set (0.003 sec)

MySQL [tomorrowday]>
```

Figure 25 EC2 to RDS database connectivity

7.3 Database Public Access Validation

The RDS database was configured with:

Public Access: No

This configuration prevents the database from receiving direct public internet connections.

In addition, the database security group only allows MySQL traffic from the web server security group.

Therefore, the intended traffic flow is:

Internet → EC2 Web Server → RDS

and not:

Internet → RDS

Screenshot

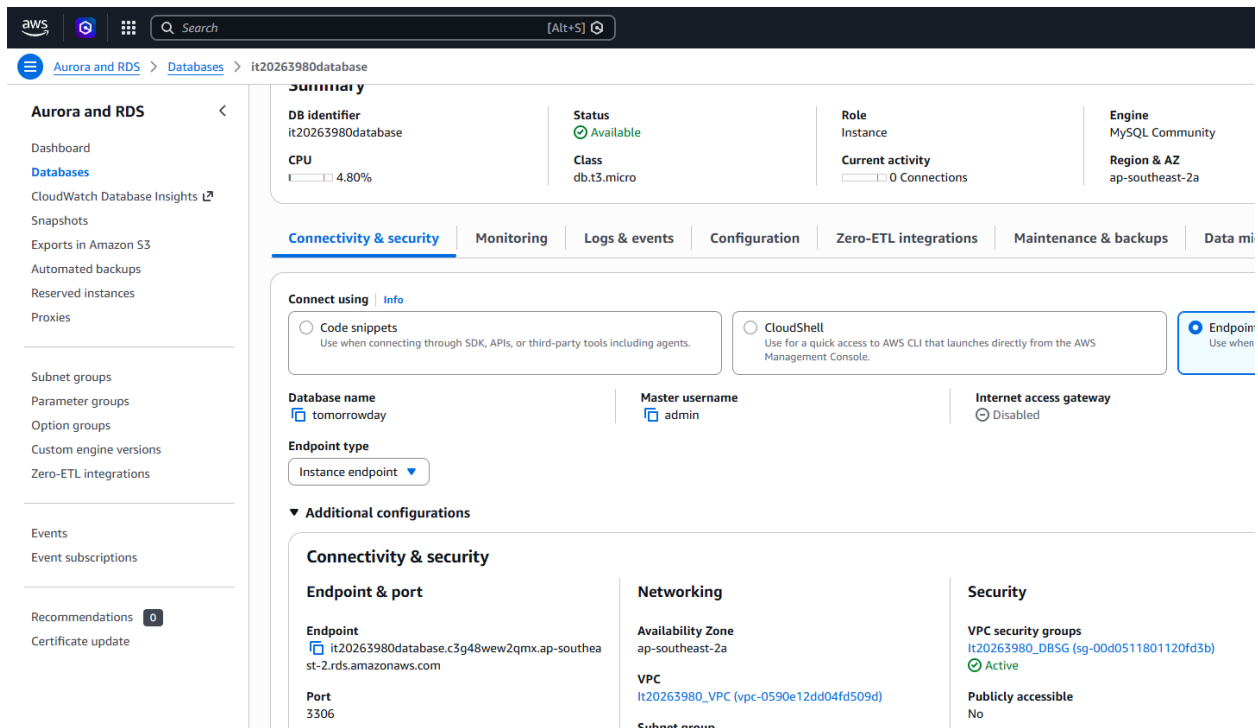


Figure 26 RDS configured without public access

This provides evidence that the database is not directly exposed to the internet.

7.4 Subnet Validation

The infrastructure was checked to confirm that each resource was deployed into the correct subnet.

EC2 Subnet Verification

The EC2 instance is deployed in:

IT20263980_PublicSubnet

Screenshot

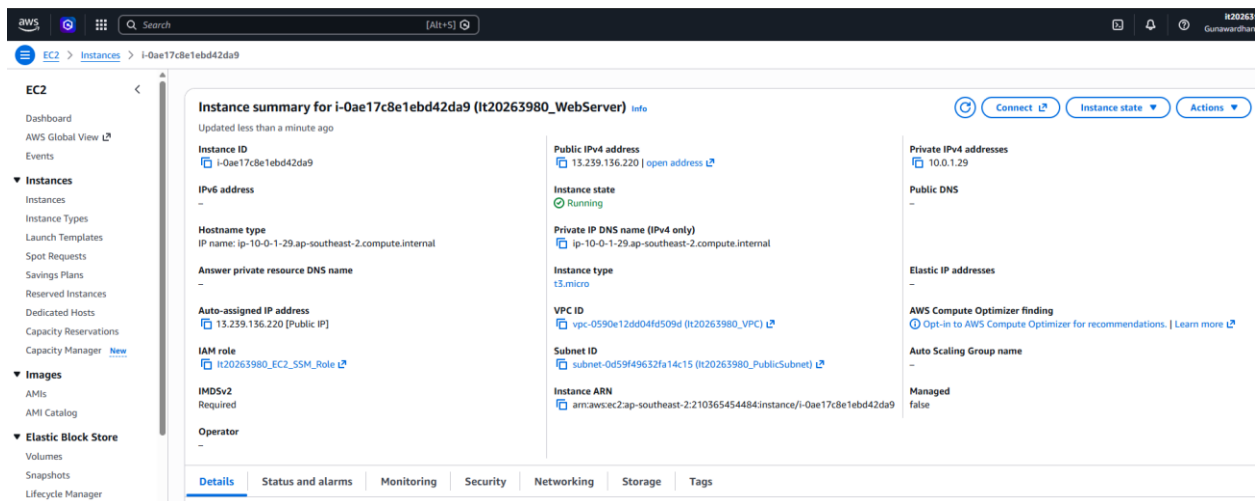


Figure 27 EC2 public subnet verification

RDS Private Subnet Verification

The RDS database is associated with the private DB subnet group.

Screenshot

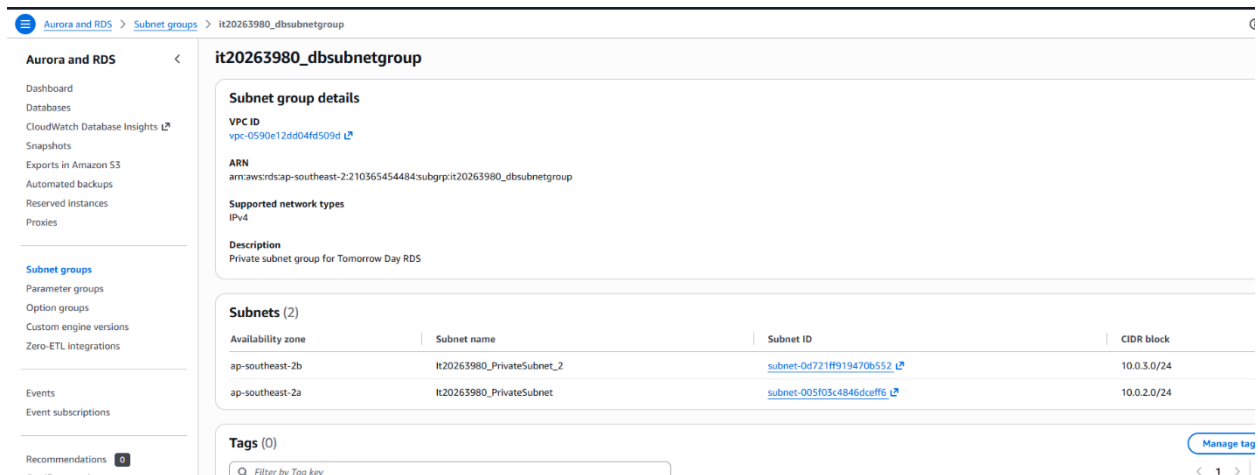


Figure 28 RDS private subnet verification

This screenshot verify that the web tier and database tier are appropriately separated.

7.5 Security Group Validation

Security group rules were reviewed to verify that only the required traffic is permitted.

Web Security Group

Expected configuration:

Rule	Port	Source
------	------	--------

HTTP	80	0.0.0.0/0
SSH	22	Administrator IP
Outbound	All	0.0.0.0/0

Database Security Group

Expected configuration:

Rule	Port	Source
MySQL	3306	IT20263980_WebSG
Outbound	All	0.0.0.0/0

Screenshot – Database Security Group Rules

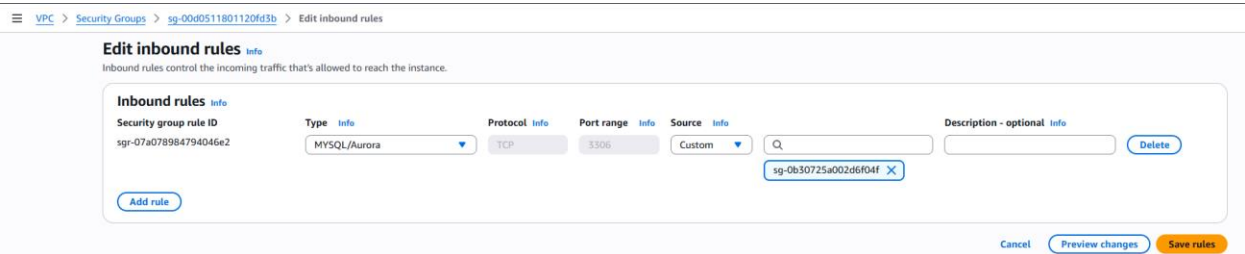


Figure 29 Database Security Group rule verification

Screenshot – Web Security Group Rules

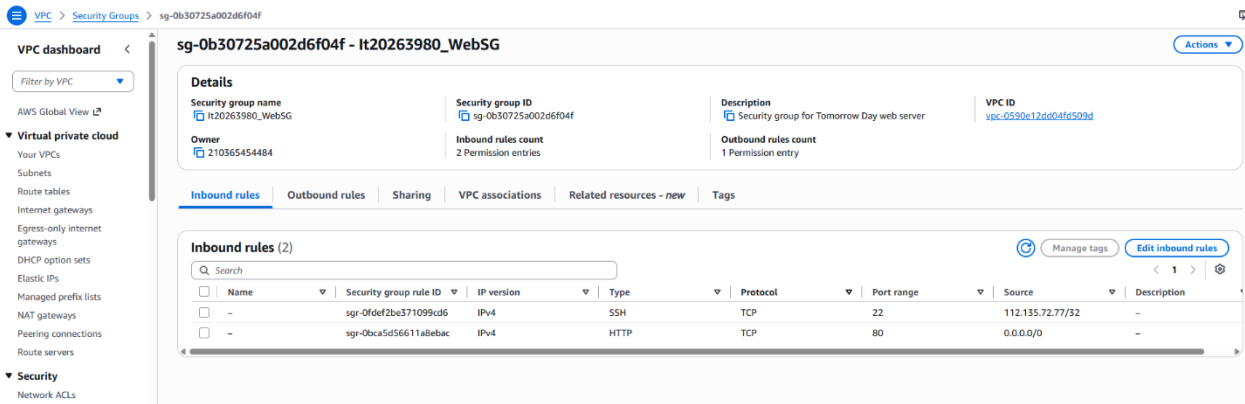


Figure 30 Web Security Group rule verification

The validation confirms that the database is not configured with an unrestricted 0.0.0.0/0 MySQL rule.

8. Security Considerations

Security was considered at multiple levels of the proposed AWS architecture.

8.1 Network Segmentation

The infrastructure separates the web and database tiers using public and private subnets.

The EC2 web server is located in the public subnet, while the RDS database uses private subnets.

8.2 Security Groups

Security groups are used to control network access.

The Web Security Group allows HTTP access from the internet and restricts SSH access to the administrator's IP.

The Database Security Group allows MySQL traffic only from the Web Security Group.

8.3 Database Isolation

The RDS database is configured with public access disabled.

This reduces the attack surface by preventing direct public access to the database.

8.4 Least-Privilege Network Access

The database security group uses the web server security group as the source rather than allowing MySQL traffic from the entire internet.

8.5 Resource Identification

AWS resources are consistently named using the Student ID:

IT20263980_

This improves resource identification and management.

8.6 Resource Tagging

Resources are tagged using project and ownership information such as:

- StudentID
- Project
- Environment
- Owner

9. Cost Considerations

The solution uses AWS managed services to reduce the requirement for physical infrastructure.

The use of Amazon EC2 and Amazon RDS allows infrastructure capacity to be provisioned according to requirements.

For the assignment implementation, small instance configurations were used where supported by the AWS Academy environment.

However, AWS resources can generate charges depending on the account type, region, resource configuration, storage, data transfer, and duration of use.

Attention should be given to:

- EC2 instance runtime
- RDS instance runtime
- RDS storage
- Public IPv4 addresses
- Data transfer
- Other provisioned AWS resources

For production deployment, cost monitoring, budgets, right-sizing, and appropriate scaling policies should be implemented.

The assignment implementation is intended as an academic cloud infrastructure demonstration rather than a complete production-scale deployment for all 12.8 million users.

10. Conclusion

This assignment proposed and implemented an AWS-based cloud solution for the Tomorrow Day business scenario.

The solution addresses the major requirements identified from the case study by replacing dependence on physical data-center infrastructure with cloud-based services.

The proposed architecture consists of:

- Amazon VPC
- Public and private subnets
- Internet Gateway
- Public EC2 web server
- Amazon RDS MySQL database
- Web Security Group
- Database Security Group

- Route Table
- CloudFormation Infrastructure as Code

The EC2 web server was successfully configured to provide web access through the internet. The RDS database was deployed without public access and was configured to accept MySQL traffic only from the web server security group.

Testing verified the main requirements, including web server accessibility, EC2-to-RDS connectivity, subnet placement, database isolation, and security group rules.

Furthermore, AWS CloudFormation was used to demonstrate Infrastructure as Code. The successful CloudFormation deployment demonstrates that the infrastructure can be defined, deployed, and managed using a repeatable template rather than relying entirely on manual configuration.

Overall, the implemented architecture demonstrates how AWS cloud services can provide a structured, secure, and manageable foundation for a digital news and business information platform.

11. References

1. Amazon Web Services. *Amazon VPC Documentation*. AWS Documentation.
2. Amazon Web Services. *Amazon EC2 Documentation*. AWS Documentation.
3. Amazon Web Services. *Amazon RDS Documentation*. AWS Documentation.
4. Amazon Web Services. *AWS CloudFormation Documentation*. AWS Documentation.
5. Amazon Web Services. *AWS Security Groups Documentation*. AWS Documentation.
6. Amazon Web Services. *AWS Academy Sandbox Environment*. AWS Academy.

Appendix A – Infrastructure as Code Using CloudFormation

```
AWSTemplateFormatVersion: '2010-09-09'
```

```
Description: >
```

```
Cloud infrastructure for Tomorrow Day  
news and business information platform.
```

```
Parameters:
```

```
StudentID:
```

```
Type: String
```

Default: IT20263980

Description: Student ID used for resource naming

AmiId:

Type: AWS::EC2::Image::Id

Default: ami-0eeab0e1473986ffd

Description: Amazon Linux 2023 AMI ID for ap-southeast-2

DBPassword:

Type: String

NoEcho: true

MinLength: 8

Description: RDS master password

Resources:

VPC

VPC:

Type: AWS::EC2::VPC

Properties:

CidrBlock: 10.0.0.0/16

EnableDnsSupport: true

EnableDnsHostnames: true

Tags:

- Key: Name

Value: !Sub '\${StudentID} VPC'

- Key: StudentID

Value: !Ref StudentID

- Key: Project

Value: TomorrowDay

- Key: Environment

Value: Assignment

Internet Gateway

InternetGateway:

Type: AWS::EC2::InternetGateway

Properties:

Tags:

- Key: Name

Value: !Sub '\${StudentID} _IGW'

- Key: StudentID

Value: !Ref StudentID

- Key: Project

Value: TomorrowDay

InternetGatewayAttachment:

Type: AWS::EC2::VPCGatewayAttachment

Properties:

InternetGatewayId: !Ref InternetGateway

VpcId: !Ref VPC

Public Subnet

PublicSubnet:

Type: AWS::EC2::Subnet

Properties:

VpcId: !Ref VPC

AvailabilityZone: ap-southeast-2a

CidrBlock: 10.0.1.0/24

MapPublicIpOnLaunch: true

Tags:

- Key: Name

Value: !Sub '\${StudentID} _PublicSubnet'

- Key: StudentID

Value: !Ref StudentID

- Key: Project

Value: TomorrowDay

- Key: Environment

Value: Assignment

Private Subnet 1

PrivateSubnet1:

Type: AWS::EC2::Subnet

Properties:

VpcId: !Ref VPC

AvailabilityZone: ap-southeast-2a

CidrBlock: 10.0.2.0/24

MapPublicIpOnLaunch: false

Tags:

```
- Key: Name
  Value: !Sub '${StudentID} PrivateSubnet 1'
- Key: StudentID
  Value: !Ref StudentID
- Key: Project
  Value: TomorrowDay
- Key: Environment
  Value: Assignment
```

```
# -----
# Private Subnet 2
# -----
```

PrivateSubnet2:

Type: AWS::EC2::Subnet

Properties:

VpcId: !Ref VPC
AvailabilityZone: ap-southeast-2b
CidrBlock: 10.0.3.0/24
MapPublicIpOnLaunch: false

Tags:

```
- Key: Name
  Value: !Sub '${StudentID} PrivateSubnet 2'
- Key: StudentID
  Value: !Ref StudentID
- Key: Project
  Value: TomorrowDay
- Key: Environment
  Value: Assignment
```

```
# -----
# Public Route Table
# -----
```

PublicRouteTable:

Type: AWS::EC2::RouteTable

Properties:

VpcId: !Ref VPC

Tags:

```
- Key: Name
  Value: !Sub '${StudentID} PublicRT'
- Key: StudentID
  Value: !Ref StudentID
- Key: Project
  Value: TomorrowDay
```

- Key: Environment
Value: Assignment

Internet Route

PublicRoute:

Type: AWS::EC2::Route

DependsOn: InternetGatewayAttachment

Properties:

RouteTableId: !Ref PublicRouteTable

DestinationCidrBlock: 0.0.0.0/0

GatewayId: !Ref InternetGateway

Public Subnet Route Table Association

PublicSubnetRouteTableAssociation:

Type: AWS::EC2::SubnetRouteTableAssociation

Properties:

SubnetId: !Ref PublicSubnet

RouteTableId: !Ref PublicRouteTable

Web Server Security Group

WebSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: Security group for Tomorrow Day web server

VpcId: !Ref VPC

SecurityGroupIngress:

- IpProtocol: tcp

FromPort: 80

ToPort: 80

CidrIp: 0.0.0.0/0

SecurityGroupEgress:

- IpProtocol: -1

CidrIp: 0.0.0.0/0

Tags:

- Key: Name

Value: !Sub '\${StudentID} WebSG'

- Key: StudentID
Value: !Ref StudentID
- Key: Project
Value: TomorrowDay
- Key: Environment
Value: Assignment

Database Security Group

DatabaseSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: Security group for Tomorrow Day RDS database

VpcId: !Ref VPC

SecurityGroupIngress:

- IpProtocol: tcp
FromPort: 3306
ToPort: 3306
SourceSecurityGroupId: !Ref WebSecurityGroup

SecurityGroupEgress:

- IpProtocol: -1
CidrIp: 0.0.0.0/0

Tags:

- Key: Name
Value: !Sub '\${StudentID}_DB-SG'
- Key: StudentID
Value: !Ref StudentID
- Key: Project
Value: TomorrowDay
- Key: Environment
Value: Assignment

RDS DB Subnet Group

DBSubnetGroup:

Type: AWS::RDS::DBSubnetGroup

Properties:

DBSubnetGroupDescription: Private subnet group for Tomorrow Day RDS

SubnetIds:

- !Ref PrivateSubnet1
- !Ref PrivateSubnet2

Tags:

- Key: Name
Value: !Sub '\${StudentID}_DBSubnetGroup'
- Key: StudentID
Value: !Ref StudentID
- Key: Project
Value: TomorrowDay
- Key: Environment
Value: Assignment

IAM Role for EC2 Systems Manager

EC2Role:

Type: AWS::IAM::Role

Properties:

RoleName: !Sub '\${StudentID}-EC2-SSM-Role'

AssumeRolePolicyDocument:

Version: '2012-10-17'

Statement:

- Effect: Allow
- Principal:
 - Service:
 - ec2.amazonaws.com
- Action:
 - sts:AssumeRole

ManagedPolicyArns:

- arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore

Tags:

- Key: StudentID
Value: !Ref StudentID
- Key: Project
Value: TomorrowDay
- Key: Environment
Value: Assignment

EC2 Instance Profile

EC2InstanceProfile:

Type: AWS::IAM::InstanceProfile

Properties:

Roles:

- !Ref EC2Role

EC2 Web Server

WebServer:

Type: AWS::EC2::Instance

Properties:

ImageId: !Ref AmiId

InstanceType: t3.micro

SubnetId: !Ref PublicSubnet

SecurityGroupIds:

- !Ref WebSecurityGroup

IamInstanceProfile: !Ref EC2InstanceProfile

Tags:

- Key: Name

Value: !Sub '\${StudentID}_WebServer'

- Key: StudentID

Value: !Ref StudentID

- Key: Project

Value: TomorrowDay

- Key: Environment

Value: Assignment

UserData:

Fn::Base64: !Sub |

#!/bin/bash

dnf update -y

dnf install -y httpd

systemctl enable httpd

systemctl start httpd

cat > /var/www/html/index.html <<EOF

<html>

<head>

<title>Tomorrow Day AWS Cloud Solution</title>

</head>

<body>

<h1>Tomorrow Day - AWS Cloud Solution</h1>

<h2>Web Server Successfully Deployed</h2>

<p>Student ID: \${StudentID}</p>

<p>Server: Amazon EC2</p>

<p>Deployment: AWS CloudFormation</p>

</body>

</html>

EOF

RDS MySQL Database

Database:

Type: AWS::RDS::DBInstance

DeletionPolicy: Snapshot

Properties:

DBInstanceIdentifier: !Sub '\${StudentID}-Database'

Engine: mysql

DBInstanceClass: db.t3.micro

AllocatedStorage: 20

StorageType: gp2

MasterUsername: admin

MasterUserPassword: !Ref DBPassword

DBName: tomorrowday

VPCSecurityGroups:

- !Ref DatabaseSecurityGroup

DBSubnetGroupName: !Ref DBSubnetGroup

PubliclyAccessible: false

BackupRetentionPeriod: 0

Tags:

- Key: Name

Value: !Sub '\${StudentID}_Database'

- Key: StudentID

Value: !Ref StudentID

- Key: Project

Value: TomorrowDay

- Key: Environment

Value: Assignment

Outputs

Outputs:

VPCId:

Description: VPC ID

Value: !Ref VPC

PublicSubnetId:

Description: Public subnet ID

Value: **!Ref** PublicSubnet

PrivateSubnet1Id:

Description: Private subnet 1 ID

Value: **!Ref** PrivateSubnet1

PrivateSubnet2Id:

Description: Private subnet 2 ID

Value: **!Ref** PrivateSubnet2

WebSecurityGroupId:

Description: Web server security group ID

Value: **!Ref** WebSecurityGroup

DatabaseSecurityGroupId:

Description: Database security group ID

Value: **!Ref** DatabaseSecurityGroup

WebServerInstanceId:

Description: EC2 Web Server Instance ID

Value: **!Ref** WebServer

WebServerPublicIP:

Description: Public IP address of web server

Value: **!GetAtt** WebServer.PublicIp

DatabaseEndpoint:

Description: RDS database endpoint

Value: **!GetAtt** Database.Endpoint.Address

DatabasePort:

Description: RDS database port

Value: **!GetAtt** Database.Endpoint.Port